

Problem 1.1: Probability density and current



The time-dependent Schrödinger equation in 3D is

$$i\hbar \frac{\partial}{\partial t} \psi(\mathbf{x}, t) = -\frac{\hbar^2}{2m} \nabla^2 \psi(\mathbf{x}, t) + V(\mathbf{x}) \psi(\mathbf{x}, t).$$

Using the Schrödinger equation, we can calculate the probability density $\rho(\mathbf{x}, t) = |\psi(\mathbf{x}, t)|^2$, which gives the probability of finding the particle at an infinitesimally small region around $\mathbf{x}$ at time t . For a closed system, the probability density should satisfy the continuity equation:

$$\frac{\partial \rho(\mathbf{x}, t)}{\partial t} + \nabla \cdot \mathbf{j}(\mathbf{x}, t) = 0.$$

This is a statement of local conservation of probability, where $\mathbf{j}(\mathbf{x}, t)$ is the probability current density (how much probability is flowing through a unit area per unit time at position $\mathbf{x}$ and time t).

- a) Derive the standard expression for the probability current density:

$$\mathbf{j}(\mathbf{x}, t) = \frac{\hbar}{2mi} (\psi^*(\mathbf{x}, t) \nabla \psi(\mathbf{x}, t) - \psi(\mathbf{x}, t) \nabla \psi^*(\mathbf{x}, t)).$$

- b) Rewrite $\mathbf{j}(\mathbf{x}, t)$ in terms of the momentum operator $\hat{\mathbf{p}} = -i\hbar \nabla$.

- c) In the Schrödinger picture and using Dirac notation, find an expression for the current operator $\hat{j}(\mathbf{x})$ in a basis-independent way. Is this operator time-dependent? What about its expectation value? (hint: recall that $\psi(\mathbf{x}) = \langle \mathbf{x} | \psi \rangle$ and $\psi^*(\mathbf{x}) = \langle \psi | \mathbf{x} \rangle$).

- d) Now consider a polar form for the wave function: $\psi(\mathbf{x}, t) = \sqrt{\rho(\mathbf{x}, t)} e^{iS(\mathbf{x}, t)/\hbar}$, where $\rho(\mathbf{x}, t)$ is the probability density and $S(\mathbf{x}, t)$ is the phase of the wave function. Show that the probability current density can be written as

$$\mathbf{j}(\mathbf{x}, t) = \frac{\rho(\mathbf{x}, t)}{m} \nabla S(\mathbf{x}, t).$$

What does this mean for a *real* wavefunction $\psi(\mathbf{x})$?

- e) For a plane-wave solution of the form $\psi(\mathbf{x}, t) = \frac{1}{\sqrt{V}} e^{i(\mathbf{k} \cdot \mathbf{x} - \omega t)}$, where V is some normalization volume (formally, $V \rightarrow \infty$ for a bona-fide plane wave), $\mathbf{k}$ is the wave vector, and ω is the angular frequency, compute the probability density $\rho(\mathbf{x}, t)$. Is it time-dependent? How about the probability current density $\mathbf{j}(\mathbf{x}, t)$?



Solution to Problem 1.1 This exercise is a straightforward application of Schrödinger's equation and some vector calculus.

- a) Compute the time derivative of the probability density:

$$\frac{\partial \rho}{\partial t} = \frac{\partial}{\partial t} |\psi|^2 = \frac{\partial}{\partial t} (\psi^* \psi) = \psi^* \frac{\partial \psi}{\partial t} + \psi \frac{\partial \psi^*}{\partial t}. \quad (1)$$

We will need Schrödinger's equation and its complex conjugate:

$$\frac{\partial \psi}{\partial t} = \frac{1}{i\hbar} \left(-\frac{\hbar^2}{2m} \nabla^2 \psi + V \psi \right), \quad (2)$$

$$\frac{\partial \psi^*}{\partial t} = -\frac{1}{i\hbar} \left(-\frac{\hbar^2}{2m} \nabla^2 \psi^* + V \psi^* \right). \quad (3)$$

Substituting these into the equation above,

$$\frac{\partial \rho}{\partial t} = \psi^* \left(\frac{1}{i\hbar} \left(-\frac{\hbar^2}{2m} \nabla^2 \psi + V\psi \right) \right) + \psi \left(-\frac{1}{i\hbar} \left(-\frac{\hbar^2}{2m} \nabla^2 \psi^* + V\psi^* \right) \right) \quad (4)$$

$$= -\frac{\hbar}{2mi} (\psi^* \nabla^2 \psi - \psi \nabla^2 \psi^*) = -\nabla \cdot \left(\frac{\hbar}{2mi} (\psi^* \nabla \psi - \psi \nabla \psi^*) \right), \quad (5)$$

proving our result that

$$\mathbf{j}(\mathbf{x}, t) = \frac{\hbar}{2mi} (\psi^*(\mathbf{x}, t) \nabla \psi(\mathbf{x}, t) - \psi(\mathbf{x}, t) \nabla \psi^*(\mathbf{x}, t)).$$

b) This is a trivial substitution using $\hat{\mathbf{p}} \rightarrow -i\hbar \nabla$.

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Okay, to be precise, we should think about how the momentum operator acts in Dirac notation:

$$\langle \mathbf{x} | \hat{\mathbf{p}} | \psi(t) \rangle = -i\hbar \nabla \langle \mathbf{x} | \psi(t) \rangle = -i\hbar \nabla \psi(\mathbf{x}, t). \quad (6)$$

where I am making clear that the operator $\hat{\mathbf{p}}$ acts on the ket $|\psi(t)\rangle$, while the “operator” ∇ acts on functions of position $\mathbf{x}$ (namely, the wave function $\psi(\mathbf{x}, t)$ in this case).

If we are okay a little bit of abuse of notation, we can write

$$\mathbf{j}(\mathbf{x}, t) = \frac{1}{2m} (\psi^*(\mathbf{x}, t) \hat{\mathbf{p}} \psi(\mathbf{x}, t) + \psi(\mathbf{x}, t) \hat{\mathbf{p}} \psi^*(\mathbf{x}, t)), \quad (7)$$

as long as we interpret this as meaning:

$$\mathbf{j}(\mathbf{x}, t) = \frac{1}{2m} (\langle \psi(t) | \mathbf{x} \rangle \langle \mathbf{x} | \hat{\mathbf{p}} | \psi(t) \rangle + \langle \psi(t) | \hat{\mathbf{p}} | \mathbf{x} \rangle \langle \mathbf{x} | \psi(t) \rangle). \quad (8)$$

c) Recall that the state $|\psi\rangle$ is basis-independent, and that when we write $\psi(\mathbf{x}) = \langle \mathbf{x} | \psi \rangle$, we are expressing the state in the position basis. One could just as well express it in the momentum basis, $\phi(\mathbf{p}) = \langle \mathbf{p} | \psi \rangle$, or any other basis for that matter. The action of some operator $\hat{O}$ on the state $|\psi\rangle$ is also basis-independent.

Using the result stated above (the careful version), we have

$$j(\mathbf{x}, t) = \frac{1}{2m} \left(\langle \psi(t) | \mathbf{x} \rangle \langle \mathbf{x} | \hat{\mathbf{p}} | \psi(t) \rangle + \langle \psi(t) | \hat{\mathbf{p}} | \mathbf{x} \rangle \langle \mathbf{x} | \psi(t) \rangle \right).$$

Technically, we are not done yet, as the questions asks for the current operator $\hat{j}(\mathbf{x})$ (note the lack of a time dependence). This is obtained by *unsandwiching* (definitely a word) the state $|\psi\rangle$:

$$\hat{j}(\mathbf{x}) = \frac{1}{2m} \left(|\mathbf{x}\rangle \langle \mathbf{x} | \hat{\mathbf{p}} + \hat{\mathbf{p}} | \mathbf{x}\rangle \langle \mathbf{x} | \right).$$

To be more precise, this *unsandwiching* operation is simply undoing the step of calculating the expectation value of the current operator, namely $j(\mathbf{x}, t) = \langle \psi(t) | \hat{j}(\mathbf{x}) | \psi(t) \rangle$.

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An important point here: the state $|\psi(t)\rangle = U(t, 0) |\psi(0)\rangle$ is time-dependent in the **Schrödinger picture**. It is obtained from the initial state by the unitary operator of time evolution $U(t, 0)$. Operators are not time-dependent in this picture (unless they have explicit time dependence).

d) You can show this by direct substitution. Sparing us both from the details, I will write the first and last step:

$$\mathbf{j}(\mathbf{x}, t) = \frac{\hbar}{2mi} (\sqrt{\rho} e^{-iS/\hbar} \nabla (\sqrt{\rho} e^{iS/\hbar}) - \sqrt{\rho} e^{iS/\hbar} \nabla (\sqrt{\rho} e^{-iS/\hbar})) = \frac{\rho}{m} \nabla S. \quad (9)$$

Note that we used the fact that $\nabla\rho = 2\sqrt{\rho}(\nabla\sqrt{\rho})$ and that $\nabla e^{iS/\hbar} = (i/\hbar)e^{iS/\hbar}\nabla S$.

Note that for a *real* wavefunction, the phase S is zero, so $\mathbf{j}(\mathbf{x}, t) = 0$ (we mean $\psi(\mathbf{x})$ is real for all $\mathbf{x}$, since the time-evolution phase does not even enter our expression for j). This implies the state is *stationary*. You already knew that from the simple observation that a real wavefunction must be an energy eigenstate with no time dependence other than a global phase.



This result is quite interesting: it shows that the flow of probability depends on the rate of change of the phase of the wave function. With an abuse of notation, we may even interpret $\nabla S/m$ as a velocity field for the probability flow, $j = \rho\mathbf{v}$. In the context of the WKB approximation, one can show that this phase is closely related to the classical action (Hamilton-Jacobi action), hence the notation S .

e) For the plane wave, the probability density is just a constant: $\rho(\mathbf{x}, t) = |A|^2$, which is independent of time, $\partial\rho/\partial t = 0$. The LHS of the continuity equation is zero, so the RHS must also be zero: $\nabla \cdot \mathbf{j} = 0$. But what is $\mathbf{j}$? Using the result above, we have that $\nabla S = \hbar\mathbf{k}$, so the probability current density is

$$\mathbf{j}(\mathbf{x}, t) = \frac{|A|^2}{m}\hbar\mathbf{k}.$$

Despite the fact that the probability density is constant in space and time, the probability density is constant flowing in the direction of the wave vector $\mathbf{k}$. This is consistent with our intuition of a plane wave representing a free particle with momentum $\mathbf{p} = \hbar\mathbf{k}$.